

RATIO, PROPORTION AND VARIATION

Ratio : Comparison of numbers by division operation. The ratio of a, b can be written as a : b. a & b are called the terms of ratio. First term “a” is called “*Antecedent*” Second term “b” is called “*Consequent*”.

$$a : b = a \div b$$
$$a : b \neq b : a$$

$$\text{Ex : } 2 : 3 = \frac{2}{3} = 0.666\dots$$

$$3 : 2 = \frac{3}{2} = 1.5$$

$$2 : 3 \neq 3 : 2$$

Simplified form of Ratio :

In the given ratio, if all the terms are not divisible by any one of the value except 1, then the terms are in simplified form. The terms of ratio must be put in simplified form only.

Ex : i) $60 : 72 : 84 = ?$ (Not in simplified form)

$$60 : 72 : 84 = 5 : 6 : 7 (\div 12)$$

ii) $85 : 119 : 153 = ?$ (Not in simplified form)

$$85 : 119 : 153 = 5 : 7 : 9 (\div 17)$$

Numbers Ratio & Numbers :

Ex : Numbers Ratio = $2 : 3 : 4$

$$\text{Numbers} = 2x, 3x, 4x$$

where “x” is any constant

(Numbers not equal to 2, 3, 4)

Types of Ratio :

1. Duplicate ratio of $a : b = a^2 : b^2$

$$\text{Duplicate ratio of } 3 : 4 = 3^2 : 4^2 = 9 : 16$$

2. Sub-duplicate ratio of $a : b = \sqrt{a} : \sqrt{b}$

$$\text{Sub-duplicate ratio of } 16 : 25 = \sqrt{16} : \sqrt{25} = 4 : 5$$

3. Triplicate ratio of $a : b = a^3 : b^3$

$$\text{Triplicate ratio of } 2 : 3 = 2^3 : 3^3 = 8 : 27$$

4. Sub-Triplicate ratio of $a : b = a^{1/3} : b^{1/3}$

Sub-Triplicate ratio of $64 : 125 = 64^{1/3} : 125^{1/3}$
 $= 4 : 5$

5. Inverse ratio of $a : b = b : a$

Inverse ratio of $3 : 5 = 5 : 3$

6. Compound ratio of $a : b, c : d, e : f, \dots$

$$= a/b \times c/d \times e/f \times \dots$$

Compound ratio of $2 : 3, 6 : 5, 10 : 7 = ?$

$$= 2/3 \times 6/5 \times 10/7$$

$$= 8/7$$

Principles of Ratio :

1st Principle of Ratio :

The terms of ratio can be multiplied or divided by the same number its value can't be changed

Ex : i) $2 : 3 = 4 : 6 (\times 2)$

ii) $3 : 5 = 15 : 25 (\times 5)$

iii) $12 : 15 = 4 : 5 (\div 3)$

2nd Principle of Ratio :

The terms of ratio must be in the same units.

Ex : i) 2hrs : 20min = ?

2hrs : 20min = 120min : 20min

$$= 6 : 1$$

i) 4 tonnes : 8 Quintals = ?

4 tonnes : 8 Quintals = 4000kgs : 800kgs

(1 Tonne = 1000kgs; 1 Quintal = 100 kgs)

$$= 5 : 1$$

3rd Principle of Ratio :

The terms of ratio must be in integers, not in fractions or decimals. If the terms are in fractions or decimal, convert them in to integers

Ex : i) $\frac{2}{3} : \frac{5}{7} = ?$

(We convert it in to integers, by doing cross multiplication)

$$2/3 : 5/7 = 2 \times 7 : 5 \times 3 = 14 : 15$$

ii) $\frac{1}{2} : \frac{3}{4} : \frac{5}{8} = ?$

(Multiply each and every term with the LCM of 2, 4, 8)

LCM of (2, 4, 8) = 8

$$\frac{1}{2} : \frac{3}{4} : \frac{5}{8} = \frac{1}{2} \times 8 : \frac{3}{4} \times 8 : \frac{5}{8} \times 8 = 4 : 6 : 5$$

iii) $0.5 : 0.25 : 0.125 = ?$

$$0.500 : 0.250 : 0.125 = 500 : 250 : 125 \\ = 4 : 2 : 1$$

Proportion :

If $a : b = c : d$, then the four terms in order a, b, c, d are in proportion.

$$a : b = c : d$$

End values “a” and “d” are called extremes, the middle values “b” and “c” are called means .

Principle of Proportion :

Product of Extremes = Product of means

$$a \times d = b \times c$$

Ex : i) $3 : 4 = 9 : 12 \rightarrow 3, 4, 9, 12$ are in proportion

$$3 \times 12 = 4 \times 9$$

ii) $12 : 15 = 4 : 5 \rightarrow 12, 15, 4, 5$ are in proportion

$$12 \times 5 = 15 \times 4$$

Continued Proportion :

If the number of ratios are equal i.e. $a : b = b : c = c : d = \dots$, then they are in continued proportion.

The terms in continued proportion are in G.P (Geometric Progression)

Properties of Proportion :**i) Invertendo Property :**

If $\frac{a}{b} = \frac{c}{d}$ then $\frac{b}{a} = \frac{d}{c}$

ii) Alternendo Property :

if $\frac{a}{b} = \frac{c}{d}$ then $\frac{a}{c} = \frac{b}{d}$

iii) Componendo & Dividendo Property :

If $\frac{a}{b} = \frac{c}{d}$ then $\frac{a+b}{a-b} = \frac{c+d}{c-d}$

Types of Proportion :i) Mean Proportion of a, b = $\sqrt{a \times b}$ Mean Proportion of 8, 18 = $\sqrt{8 \times 18} = \sqrt{144} = 12$ ii) Third proportional of a, b = b^2/a Third proportional of 3, 6 = $6^2/3 = 36/3 = 12$ iii) Fourth proportional of a, b, c = $\frac{b \times c}{a}$ Fourth proportional of 4, 8, 12 = $(8 \times 12)/4 = 24$ **Variation :****i) Direct variation :**x, y ----- $\rightarrow$ "x" increases then "y" also increases

"x" decreases then "y" also decreases

 $X \propto y$ ---- $\rightarrow x = k \times y$, where "k" is proportionality

Constant

 $\frac{X}{Y} = k$, Ratio is constant**ii) Inverse variation :**

x, y ----- "x" increases then "y" decreases

"x" decreases then "y" increases

 $X \propto 1/y$ ---- $x = k \times 1/y$, where "k" is proportionality Constant $X \times Y = \text{Constant}$

Product = Constant

iii) Combined Variation :x, y & z ---- $\rightarrow x \propto y/z$ ---- $\rightarrow x = k \times y/z$ $\frac{X \times Z}{Y} = \text{Constant}$

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**COMPANIES SPECIFIC QUESTIONS
All models – Sort Cuts**

1. In the ratio 3 : 7, if the antecedent is 45, then find the Consequent ?

$$3 : 7 = 45 : ?$$

$$3 \times 15 = 45$$

$$7 \times 15 = 105 = \text{Consequent}$$

2. If $a : b = 2 : 3$ and $b : c = 4 : 5$, then find $a : b : c$?

Short - Cut

$$\begin{array}{ccc} 2 & : & 3 \\ \downarrow & \nearrow & \downarrow \\ 4 & : & 5 \end{array}$$

$$2 \times 4 : 4 \times 3 : 3 \times 5 = 8 : 12 : 15$$

3. If $a : b = 4 : 3$, $b : c = 5 : 2$, then find $a : c$?

Short - Cut

$$\begin{array}{ccc} 4 & : & 3 \\ \downarrow & : & \downarrow \\ 5 & : & 2 \end{array}$$

$$4 \times 5 : 3 \times 2 = 20 : 6 = 10 : 3$$

4. If $2A = 3B = 4C$, then find $A : B : C$?

Short - Cut

$$A : B : C = 3 \times 4 : 2 \times 4 : 2 \times 3 = 12 : 8 : 6 = 6 : 4 : 3$$

5. If $a : b = 2 : 3$, $b : c = 4 : 5$ and $c : d = 6 : 7$, then find

i) $a : b : c : d$

ii) $a : d$

iii) $b : d$

Short - Cut

$$\begin{array}{ccc} 2 & : & 3 \\ \downarrow & \nearrow & \downarrow \\ 4 & : & 5 \\ \downarrow & \nearrow & \downarrow \\ 6 & : & 7 \end{array}$$

$$\begin{aligned} a : b : c : d &= 2 \times 4 \times 6 : 4 \times 6 \times 3 : 6 \times 3 \times 5 : 3 \times 5 \times 7 \\ &= 48 : 72 : 90 : 105 \end{aligned}$$

i) $a : b : c : d = 48 : 72 : 90 : 105$

ii) $a : d = 48 : 105$

iii) $b : d = 72 : 105 = 24 : 35$

6. If $\frac{A}{2} = \frac{B}{3} = \frac{C}{4}$, then find $A : B : C$?

Short Cut

$$2 : 3 : 4$$

7. If $a : b = 3 : 4$, then find $(2a+b) : (2a-b)$?

$$a : b = 3 : 4 \rightarrow a, b = 3k, 4k$$

$$(2 \times 3k + 4k) : (2 \times 3k - 4k) = 10 : 2 = 5 : 1$$

8. If $a : b = 3 : 4$, then find $(2b+b+1) : (2a-b+1)$?

Can't be determined

9. If $a+b : a-b = 5 : 1$, then find $a : b$?

Short Cut

$$5+1 : 5-1 = 6 : 4 = 3 : 2$$

10. The two numbers are in the ratio $3 : 7$ and the smaller number is 150, then find the larger number ?

Short Cut

$$3 \rightarrow 150 \quad (3 \times 50)$$

$$7 \rightarrow ? = 7 \times 50 = 350$$

11. In a college, the number of Boys and Girls are in the ratio $5 : 3$ and the strength of the college is 480, then find the number of Girls ?

Short Cut

$$8 \rightarrow 480 \quad (5+3 = 8)(8 \times 60)$$

$$3 \rightarrow ? = 3 \times 60 = 180$$

12. The prices of Scooter and Car are in the ratio $3 : 7$, the price of car is Rs.80000 more than the price of Scooter. Find the price of Scooter ?

Short Cut

$$4 \rightarrow 80000 \quad (7-3 = 4)$$

$$3 \rightarrow ? = 3 \times 20000 = 60000$$

13. Rs.375 is divided among A, B and C in such a way that, if each of them had received Rs.25 less their shares would have been in the ratio $2 : 3 : 5$. Find A's share ?

$$2 : 3 : 5 \rightarrow 300 \quad (375 - 75 = 300)$$

$$10 \rightarrow 300$$

$$2 \text{ -----} \rightarrow ? = 2 \times 30 = 60$$

$$A's \text{ Share} = 60 + 25 = 85$$

14. The ratio between first and Second number is 2 : 3, the ratio between Second and third number is 5 : 2 and the first number is 40, then find the 3rd number ?

$$\begin{array}{ccc} 2 & : & 3 \\ \downarrow & \nearrow & \downarrow \\ 5 & : & 2 \end{array}$$

$$F : S : T = 10 : 15 : 6$$

$$10 \text{ -----} \rightarrow 40 \quad (10 \times 4 = 40)$$

$$6 \text{ -----} \rightarrow ? = 6 \times 4 = 24$$

15. A person divides his property, so that his son's share to wife, wife's share to daughter, both in the ratio 3 : 1. If daughter's share is Rs.80000 less than Son's share, then find the total worth of the property ?

$$S : W = 3 : 1$$

$$W : D = 3 : 1$$

$$3 : 1$$

$$3 : 1$$

$$S : W : D = 9 : 3 : 1$$

$$8 \text{ -----} \rightarrow \text{Rs.}80000 \quad (9-1=8)$$

$$13 \text{ -----} \rightarrow ? = 130000$$

16. A bag contains an equal number of Rs.1, 50paise and 25paise coins. If the total value in the bag is Rs.35, then find the number of coins of each type ?

100p	50p	25p
No. of Coins = X	X	X
Total value, $100X + 50X + 25X = 3500$		
$175X = 3500$		
$X = 20$		

17. A bag contains denominations of Rs.10, Rs.20 and Rs.50 coins are in the ratio 3 : 2 : 1, the total value in the bag is Rs.1200. Find the number of Rs.10 notes ?

Rs.10	Rs.20	Rs.50
3x	2x	1x
Total value, $10 \times 3x + 20 \times 2x + 50 \times 1x = 1200$		
$120x = 1200$		
$X = 10$		
Number of Rs.10 notes = $3 \times 10 = 30$		

18. 40L mixture contains Milk and Water in the ratio 3 : 2 . If 5 litres of water is added to the mixture, then what is the ratio of Milk and Water ?

$$\begin{aligned}
 M : W &= 3 : 2 \longrightarrow 40L \\
 5 &\longrightarrow 40L \quad (\times 8) \\
 M &= 3 \longrightarrow 24L \quad (\times 8) \\
 W &= 2 \longrightarrow 16L \quad (\times 2) \\
 M : W &= 24 : 16+5 \\
 &= 24 : 21 = 8 : 7
 \end{aligned}$$

19. A mixture contains Spirit and Alcohol in the ratio 5 : 4. If 10 litres of Alcohol is added to the mixture, then their quantities becomes equal. What is the quantity of Spirit ?

$$\begin{aligned}
 5 &: 4 \\
 &\quad \downarrow 1 \\
 5 &: 5 \\
 1 &\longrightarrow 10L \\
 5 &\longrightarrow ? = 50L
 \end{aligned}$$

20. The two numbers are in the ratio 3 : 4. If each is increased by 10, the ratio changes to 4 : 5. Find the largest number ?

$$\begin{aligned}
 3 &: 4 \\
 1 &\downarrow \quad 1 \downarrow \\
 4 &: 5 \\
 1 &\longrightarrow 10 \\
 4 &\longrightarrow ? = 40 \text{ (In 3:4, larger)}
 \end{aligned}$$

21. The two numbers are in the ratio 3 : 5, if each is increased by 10, the ratio changes to 2 : 3. Find the sum of the two numbers ?

$$\begin{aligned}
 3 &: 5 \\
 1 &\downarrow \quad 1 \downarrow \\
 2 &: 3 = 4 : 6 \quad (\times 2) \\
 1 &\longrightarrow 10 \\
 8 &\longrightarrow ? = 80
 \end{aligned}$$

22. The incomes of A and B are in the ratio 5 : 6 and each saves Rs.1000. If their expenses are in the ratio 4 : 5, then find A's income ?

$$\begin{aligned}
 5 &: 6 \\
 1 &\downarrow \quad 1 \downarrow \\
 4 &: 5 \\
 1 &\longrightarrow 1000 \\
 5 &\longrightarrow ? = 5000 \\
 \text{(Saving = Income – Expenditure)}
 \end{aligned}$$

23. The incomes of P and Q are in the ratio 5 : 7 and each saves Rs.500. If their expenses are in the ratio 9 : 13, then find B's expenditure ?
Terms of Incomes ratio must be greater than savings ratio.

$$\begin{array}{rcccl}
 5 & : & 7 & = & 10 & : & 14 \\
 & & & & \downarrow & & \downarrow \\
 & & & & 1 & & 1 \\
 & & & & 9 & : & 13 \\
 1 & \text{-----} & \rightarrow & 500 \\
 B = 14 & \text{-----} & \rightarrow & ? = 7000 \text{ (Income)} \\
 B's \text{ expenditure} & = & 7000 - 500 = 6500
 \end{array}$$

24. Two glasses contain Milk and water in the ratio 2 : 1 and 3 : 1, if these two are mixed together then what is the ratio of Milk and Water ?

$$M : W = \left(\frac{2}{3} + \frac{3}{4}\right) : \left(\frac{1}{3} + \frac{1}{4}\right) = 17 : 7$$

25. Two vessels contain Spirit and Alcohol in the ratio 2 : 1 and 3 : 1, if these two are mixed in the ratio 3 : 2, then what is the ratio of Milk and Water ?

$$M : W = \left(\frac{2}{3} \times 3 + \frac{3}{4} \times 2\right) : \left(\frac{1}{3} \times 3 + \frac{1}{4} \times 2\right)$$

$$= \left(2 + \frac{3}{2}\right) : \left(1 + \frac{1}{2}\right)$$

$$= \frac{7}{2} : \frac{3}{2} = 7 : 3$$

26. Which least number must be added to the numbers 3, 4, 11, 14 so that they are in proportion ?

- a)1 b)2 c)3 d)4

If Product of extremes = Product of means, then only the terms are in proportion.

$$3, 4, 11, 14 \text{ -----} \rightarrow \text{not in proportion}$$

$$+1 \text{-----} 4, 5, 12, 15 \text{ -----} \rightarrow \text{are in proportion}$$

27. $x \propto y$, If $x = 20$, then $y = 40$. Find X , if $y = 10$?

$$x/y = k$$

$$X1/y1 = x2/y2$$

$$20/40 = x/10$$

$$X = 5$$

28. $x \propto \sqrt{y}$, If $x = 10$, then $y = 20$. Find y , if $x = 20$?

$$X^2/y = k$$

$$10^2/20 = 20^2/y$$
$$Y = 80$$

29. $x \propto y^{1/3}$, If $x = 10$, then $y = 20$. Find y , if $x = 20$?

$$X^3/y = k$$

$$10^3/20 = 20^3/y$$

$$Y = 160$$

30. $x \propto 1/y^{1/3}$, if $x = 10$, then $y = 20$. Find y , if $x = 20$?

$$X^3y = k$$

$$10^3 \times 20 = 20^3 \times y$$

$$Y = 2.5$$



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Companies Specific Questions for Practice

1. In a bag, there are coins of 25 p, 10 p and 5 p in the ratio of 1 : 2 : 3. If there is Rs. 30 in all, how many 5 p coins are there?

- a)150 b)120 c)140 d)160

2. Two numbers are in the ratio 3 : 5. If 9 is subtracted from each, the new numbers are in the ratio 12 : 23. The smaller number is:

- a)12 b)10 c)33 d)15

3. If 40% of a number is equal to two-third of another number, what is the ratio of first number to the second number?

- a)1 : 2 b)2 : 3 c)3 : 4 d)5 : 3

4. The salaries A, B, C are in the ratio 2 : 3 : 5. If the increments of 15%, 10% and 20% are allowed respectively in their salaries, then what will be new ratio of their salaries?

- a)23 : 33 : 60 b)20 : 30 : 60 c)23 : 35 : 60 d)10 : 13 : 30

5. If Rs. 782 be divided into three parts, proportional to $\frac{1}{2} : \frac{2}{3} : \frac{3}{4}$, then the first part is:

- a)102 b)52 c)204 d)208

6. The sum of three numbers is 98. If the ratio of the first to second is 2 : 3 and that of the second to the third is 5 : 8, then the second number is:

- a)20 b)30 c)45 d)15

7. Salaries of Ravi and Sumit are in the ratio 2 : 3. If the salary of each is increased by Rs. 4000, the new ratio becomes 40 : 57. What is Sumit's salary?

- a)36000 b)32000 c)30000 d)34000

8. The ratio of the number of boys and girls in a college is 7 : 8. If the percentage increase in the number of boys and girls be 20% and 10% respectively, what will be the new ratio?

- a)20 : 21 b)18 : 19 c)21 : 22 d)17 : 18

9. Present age of Vinod and Ashok are in ratio of 3:4 respectively. After 5 years, the ratio of their ages becomes 7:9 respectively. What is Ashok's present age is ?

- a)40 yrs b)50 yrs c)30 yrs d)45 yrs

10. At present, the ratio between ages of Ram and Shyam is 6:5 respectively. After 7 years, Shyam's age will be 32 years. What is the present age of Ram?

a)20 yrs

b)25 yrs

c)30 yrs

d)40 yrs

11. The present ages of A, B and C are in proportions 4:5:9. Nine years ago, sum of their ages was 45 years. Find their present ages in years

a)15, 19, 35

b)17, 21, 37

c) 18, 20, 38

d)16, 20, 36

12. The ratio 5:3 represents 16 liters of a mixture containing milk and water. If 4 liters of water is added and 4 liters of milk is extracted from the mixture, then the ratio of the mixture will be:

a)3 : 5

b) 2 : 3

c) 1 : 2

d) 4 : 5

13. If the ages of Jacob, Max and Samuel are in the proportion 3:5:7 and the average of their ages is 25 years, then find the age of the youngest person ?

a)12 yrs

b)15 yrs

c)14 yrs

d)18 yrs

14. The ratio of the speed of two trains is 7:8. If the second train covers 400 km in 4 h, find out the speed of the first train.

a)80km/hr

b)85km/hr

c)87.5 km/hr

d)100km/hr

15. Syrup and Water are mixed in 2:1 ratio to form 60 litres of a mixture. How much water needs to be added to make the ratio 1:2 ?

a)60 L

b)40 L

c)50 L

d)30 L

16. The continued ratio of Rs.200, Rs.400 and Rs.600 is

a)2 : 4

b)4 : 6

c)6 : 2

d)2 : 4 : 6

17. The marks scored by a student in three subjects are in the ratio of 4 : 5 : 6. If the candidate scored an overall aggregate of 60% of the sum of the maximum marks and the maximum marks in all three subjects is the same, in how many subjects did he score more than 60%?

a)1

b)2

c)3

d)None of the subjects

18. The ratio of marks obtained by vinod and Basu is 6 : 5. If the combined average of their percentage is 68.75 and their sum of the marks is 275, find the total marks for which exam was conducted.

a)150

b)200

c)400

d)None of these

19. The monthly salaries of two persons are in the ratio of 4 : 7. If each receives an increase of Rs.25 in the salary, the ratio is altered to 3 : 5. Find their respective salaries.

a)120 & 210

b)80 & 140

c)180 & 300

d)200 & 350

20. A, B and C play cricket. The ratio of the runs scored by A and B is 3 : 2. The ratio of the runs scored by B and C is 3 : 2. Together, they score 342 runs. How many runs did A make?

a)162

b)108

c)72

d)None of these

KEY

1.a	2.c	3.d	4.a	5.c
6.b	7.d	8.c	9.a	10.c
11.d	12.a	13.b	14.c	15.a
16.d	17.a	18.b	19.d	20.a

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